Sales and Service mix optimization - A primary data analysis

A Proposal report for the BDM capstone Project

Submitted by

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1. Executive Summary & Problem Context

Modern consumer electronics, commercial hardware, and office automation businesses face severe margin compression in core product retail. Traditional retail strategies focus almost exclusively on hardware unit throughput. However, rising customer acquisition costs, aggressive e-commerce discounting, and inventory carrying overhead have compressed pure product sales margins down to 7–12%. In contrast, post-warranty after-sales services, including Annual Maintenance Contracts (AMC), preventative checkups, and priority breakdown repairs, command gross contribution margins exceeding 50% and provide consistent, non-cyclical cash flow.

Despite the superior unit economics of after-sales services, dealerships frequently struggle to establish the optimal operating mix. Over-indexing on hardware sales drains working capital and crowds out technician bandwidth, while over-indexing on maintenance services without an adequate hardware installed base leads to technician underutilization and customer attrition. This project proposes a data-driven primary investigation to empirically determine the optimal resource and revenue mix between product sales and service contracts.

2. Research Objectives & Key Questions

This capstone research addresses three specific empirical questions:

- **Primary Research Question 1:** What are the primary consumer and business drivers that dictate customer willingness to enroll in structured after-sales maintenance agreements (AMC / extended warranty) at the point of sale?
- **Primary Research Question 2:** How do unit contribution margins, customer retention rates, and Customer Lifetime Value (CLV) vary between pure transactional buyers versus service-contracted accounts?
- **Primary Research Question 3:** Under finite technician labor hours, retail floor capital, and customer turnaround time constraints, what is the mathematically optimal sales-to-service ratio that maximizes net enterprise contribution margin?

3. Theoretical Framework & Literature Grounding

This study is anchored on the seminal theory of *Servitization* (Vandermerwe & Rada, 1988; Baines et al., 2009), which posits that manufacturing and retail organizations achieve sustainable competitive advantages by bundling physical goods with value-added service solutions. Furthermore, according to Reichheld & Sasser's retention economics, increasing customer retention rates by 5% increases profits by 25% to 95%, primarily driven by the lower cost of servicing existing accounts relative to new acquisition. This project operationalizes these concepts into a practical Linear Programming (LP) optimization model.

4. Primary Data Collection Strategy & Survey Instrument

To capture realistic customer preferences and purchase behaviors, a structured primary field survey will be administered across 220 commercial buyers and retail consumers across Tier-1/Tier-2 urban multi-brand IT and equipment service centers. The sampling strategy uses stratified sampling across three core customer cohorts:

Target Cohort	Sampling Frame	Target Sample	Data Capture Focus
B2C Retail Consumers	Retail walk-ins & home appliance buyers	110 (50%)	Price sensitivity, warranty perception, DIY tendency
SME Commercial Clients	Small businesses, clinics, legal/accounting firms	75 (35%)	Downtime tolerance, AMC adoption, turnaround speed
Enterprise Accounts	Mid-to-large corporate branch offices	35 (15%)	SLA commitments, multi-year contracts, fleet scale

5. Proposed Analytical Modeling & Methodological Pipeline

The research executes a rigorous four-stage analytical pipeline: **(1) Data Scrubbing & Validation:** Outlier detection, handling skewness, normality verification. **(2) Exploratory Data Analysis (EDA):** Bivariate correlation between turnaround times, satisfaction (CSAT), and AMC adoption. **(3) Econometric Modeling:** Binary logistic regression modeling the odds of service contract opt-in. **(4) Constrained Mathematical Optimization:** Formulating an integer Linear Program (LP) solved via the Simplex method to identify the exact profit-maximizing sales-service bundle mix.

6. Project Timeline & Capstone Milestones

Milestone Phase	Scheduled Weeks	Key Deliverables	Status
Phase 1: Proposal	Weeks 1 - 3	Problem statement, literature review, questionnaire draft	Submitted
Phase 2: Mid-Term	Weeks 4 - 7	Field survey execution (N=220), data scrubbing, EDA	Scheduled
Phase 3: Final Report	Weeks 8 - 11	Econometric models, LP optimization, strategic insights	Scheduled
Phase 4: Final Viva	Week 12	Executive presentation deck & oral defense	Scheduled